

- STAT 20 SUMMER 2026

5. Imagine that you wanted to predict the average speed for three flights: one in which a plane travels 10 miles, one in which a plane travels 500 miles and another in which the plane travels 5000 miles. Which of these three predictions, if any, do you expect to be accurate, and why? Comment on each of the three potential predictions separately (use at least three sentences).
6. Remake your scatter plot but add a line representing your linear model by adding a layer with `geom_smooth(method = "lm")`. Sketch the new visualization below. How well does the line fit the data? Answer in one to two sentences.
7. How could you modify your model to improve the predictions that it makes? Answer in one to two sentences.